

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Midterm
Semester: Spring 2024

Duration: 1 Hour 15 Min
Full Marks: 25

CSE 470: Software Engineering

SET B

Name: *f*

ID: *0000000000*

Section: *10*

Q1. During the winter trip to climb Mount Everest, Bob and Alice spoke with their trainer and noted everything about all the checkpoints on a sheet of paper. They maintained another sheet of paper for important points at each checkpoint. At the end of each day, they set up camp and discussed their achievements for that day, along with the challenges they might face the next day and those they encountered that day, as well as the distance they covered. After reaching each checkpoint, they held a feast, and shared their achievement with other climbers. They also maintained a map to track their progress. This continued until they reached the peak of Mount Everest.

After the winter trip, Alice and Bob were given the task of developing a Learning Management System (LMS). Here, learners can enroll in courses based on their job roles. Instructors can upload educational videos or articles. Learners can attend monthly quizzes to test their knowledge. Alice and Bob need to ensure that the application runs smoothly on both old and new devices. Most importantly, user data should be secured from different cyber attacks, and the application should be easy to update with new features in the future. For the development of this application, they decided to follow the methodology they used during their winter trip.

- A. What methodology will be followed for the LMS application? [Mark 1] [CO1]
- B. Explain the detailed implementation process of the chosen methodology. [Mark 4] [CO1]
- C. Find out 2 functional and non-functional requirements from the given scenario. [Mark 2] [CO1]

Q2. After completing the CSE470 - Software Engineering course, your friends, now software development experts, are launching a SaaS startup focused on songwriters and composers, based on the Large Language Model (LLM). They're really enthusiastic about releasing the whole application to end-users as fast as possible. Their aim is to create a user-friendly UI. Upon logging in, users access a dashboard to search songs by genre, view prompts, and navigate to the song generation page. Here, users can input prompts for lyric generation with options to regenerate or input new prompts. Additional features include generating instrumentals, earning money by sharing songs, and a collaboration page for joint song creation. Users can publish their songs to Spotify or Apple Music. A feedback form will allow users to report bugs or suggest ideas, with plans to address issues and integrate user suggestions in future updates. Following the Verification and Validation model ensures a thorough approach to testing and refining the application.

- A. Do you think the verification and validation model would be suitable for the scenario? What SDLC model would be suitable? Explain your chosen SDLC model. [Mark 1+1+4] [CO1]

Q3. Long ago, there were some nations that lived together in harmony. They were always ready to do anything to live together happily. They protected each other using their own power. Fire nations who represented life and energy with fire, bodyguard, weapon and their powers were manipulation of fire, emphasizing power and aggression. Another nation was earth who were famous for rocks and stability using earth, weapons and manpower. Their powers were Control over earth, symbolizing strength and endurance. Besides, the Water Tribes guardians of the world's balance, wielded the power of waterbending, a skill that epitomized adaptability and healing, harnessing the fluid essence of water to protect and sustain life across the globe. Last but not least, the Air Nomads were renowned for their mastery of airbending, a practice that cherished freedom and harmony, skillfully channeling the element of air to navigate and preserve peace within the world. And finally, there was another nation that was the master of all those nations and could bend all those elements, tasked with maintaining balance and peace were called avatars. But everything changed when the Fire Nation invaded. Only the Avatar master of all four elements, can stop them. But when the world needs him most, he vanishes. A hundred years later, the new Avatar was discovered, an airbender named Aang. Together they must help Aang master the elements and save the world.

A. Draw the Class Diagram according to the above scenario. [Mark 5] [CO2]

Q4. "GroceryBD" is one of the most popular grocery store chains in the country.

A: Five years ago, when the company just started its journey with only one store in Dhanmondi 9A, it used software with which one could only check if an item was in stock. The software was built following the MVC architecture.

B: GroceryBD has been able to expand the scope of its business in recent years. They currently have 20 outlets in Dhaka. So, the quality of their software has improved too. Now one can check if any item is in stock at the other outlets too. Apart from checking products in stock, the whole purchasing and billing process is done using the software. Customers can also buy products online without the hassle of physically coming to the store. Hence, different security measures have been adopted for online payments. Also, there is a chat option on the website for customers to make any inquiries or complaints. Different teams are working on different sectors to make the website more robust and user-friendly.

In the website, a customer can also upload an image of the item they want to purchase. The software will collect the provided image, crop it so that the object is in the center, resize it to 32*32 pixels, sharpen the image, and then match it with all the images present in the database. If any image matches, the customer will be redirected to the corresponding item's page.

- A. In situation A, do you think using MVC architecture was the optimal choice? Justify your answer. [Mark 1] [CO3]
- B. Which software architecture/architectures would be suitable for developing the software in the current time (situation B)? [Mark 4] [CO3]
- C. How to stage/add all of your code and after that how to commit your code with the message "first commit done"? In Addition how to push your code to "Dev1Branch"? [Mark 2] [CO3]